

## **Use-Case Flow**

**Use Case:** UC-03 — Reserve Equipment

**Primary Actor:** Student

### **Preconditions**

1. Student is authenticated.
2. Equipment is available for reservation.
3. Equipment has valid calibration status.

### **Main Success Scenario**

1. Student selects the equipment.
2. System displays available time slots.
3. Student selects a date and time slot.
4. System verifies equipment calibration.
5. System checks for overlapping reservations.
6. System validates the reservation duration.
7. System creates the reservation.
8. System displays the reservation confirmation.

### **Alternate Flow — Equipment Unavailable**

- 5a.** System detects an overlapping reservation.
- 5b.** System rejects the reservation request.
- 5c.** System displays an unavailable message.
- 5d.** System displays alternative available time slots.
- 5e.** Student selects another available slot.

### **Postconditions**

1. Reservation is recorded successfully.
2. Equipment is reserved for the selected time slot.
3. Student receives confirmation.